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Part-01 Introduction

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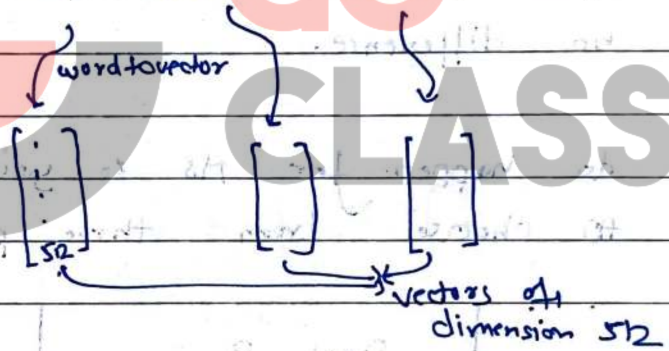
Linear Algebra

Need of Linear Algebra.

→ Linear Algebra is a tool to deal with matrices.
It is present & used everywhere because data is everywhere. & every data is matrix.

→ It is a fuel of ML.

→ Any data can be represented in Matrix.
Word Embedding: Representing a word with a vector
Ex: I am Karan.



It also records the semantics of the word
Ex: vector of (king − man + woman) = vector of queen.

This is not possible via ascii values.
It is used in ML. So from this we get the need of Linear Algebra.

→ Another use case of LA is in recommendation engines.

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AIR 1 of 2017 batch chose M
So that he can spend more time at IISC.

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Many industries use LA :
- Statistics
- Chemical Physics
- Genomics
- Robotics
- Word Embeddings
- Image Processing
- Quantum Physics

☆ Linear Algebra for GATE
↳ Atleast 1 Question (2-3 Marks) in Exam
↳ Also important for Interview if not enough good rank.

~~Adv~~

* Placements & Higher Studies options are very exactly the same after MTech and MS (or MTech Research). There is no difference.

Interviews do happen for MS & you are offered to choose from three pools of subjects

Pool A	Pool B	Pool C
Theory	System	Intelligence
Algo, DS, Graph Theory	70% OS, COA, Compiler	70% LA, Probability Calculus

Linear Algebra + Probability
⇓
Machine Learning.

Direction → Yes → Vector
→ No → Scaler.

★ **Scaler:** Any number is scaler. Ex. 1

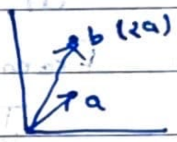
★ **Vector:** Collection of scalers. Ex. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ R^2

Like in 2D or 3D coordinate Systems.

R^n means vector having n coordinates are scalers.

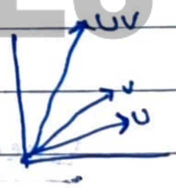
★ Scalar times vector

$$2 \times \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a} = \underbrace{\begin{bmatrix} 2 \\ 4 \end{bmatrix}}_{b}$$



★ Vector Addition

$$\underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{U} + \underbrace{\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}}_{V} = \underbrace{\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}}_{UV}$$



★ Linear Combination of Vectors

$$3 \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

↳ Linear Combination of $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ & $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

→ Finding Linear Combination of coefficient & vectors is very Easy. (10th Class thing)
But flip of the question i.e. to find the coefficient if we are given vectors & linear combination of them is not so easy. (GATE thing)

" This gentle introduction to scalars and vectors seems simple, but you may be surprised to learn that nearly all of LA is built up of scalars & vectors.
From humble beginnings, amazing things emerge. Just think of everything you can build from wood planks and nails "
↳ Some famous book author.

★ Linearly Dependent Vectors

Ex: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ & $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ easy to check if only two vectors

So by multiplying one vector with some scaler if we get another vector then it is linearly dependent. else independent.
It will become hard to see if set contains more than two vectors. (will need a method to check & will not be able to do via intusion)

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Imp ★ Independence is a property of set of vectors.
i.e. a set of vectors can be linearly independent or dependent; it doesn't makes sense to ask whether a single vector; or as vector within a set is independent.

Example of linearly dependent vector set

Ex:1

$$\left\{ \underset{u}{\begin{bmatrix}1\\2\\3\end{bmatrix}}, \underset{v}{\begin{bmatrix}2\\4\\7\end{bmatrix}}, \underset{w}{\begin{bmatrix}3\\6\\10\end{bmatrix}} \right\}$$

only one is required for linear independent

because $\underline{u+v=w}$, or $v=w-u$ or $u=w-v$

Ex.2

$$\left\{ \underset{u}{\begin{bmatrix}1\\2\\3\end{bmatrix}}, \underset{v}{\begin{bmatrix}2\\4\\7\end{bmatrix}}, \underset{w}{\begin{bmatrix}3\\6\\9\end{bmatrix}} \right\}$$

because $\underline{3u+0v=w}$ or $u=\frac{1}{3}w+0v$
↳ ∴ Coefficients can be zero (any real number)

Imp. ★ correct
Any set is linear dependent if ~~any~~ atleast one vector can be represented as linear combination of other vectors.

Ex:3

$$\left\{ \underset{u}{\begin{bmatrix}1\\2\\3\end{bmatrix}}, \underset{v}{\begin{bmatrix}2\\4\\9\end{bmatrix}}, \underset{w}{\begin{bmatrix}0\\0\\0\end{bmatrix}} \right\}$$

It is LD set as $\underline{w=0u+0v}$

Imp.

★ → from Ex 3, we can say that "a set containing zero vector is always Linearly Dependent."

There can be multiple set of coefficients possible for linear combination of vector.

Que: If

$$V_1 = C_2V_2 + C_3V_3 \dots + C_nV_n$$

then,

Can I represent V_2 as Linear Combination of other vectors?

Ans: Yes if C_2 is not zero then we can represent V_2 as LC of other vectors.

Que: T/F: If a set of vectors are linearly Dependent then, there is atleast one vector which can be represented by a LC of other vectors.

Ans: Always True. That is the definition of Linear Dependency.

GATE PYQ

Que: T/F: If a set of vectors are Linearly Dependent then all vectors be represented by a linear combination of other ~~vectors~~ vectors.

Ans: False, Not all can be represented (specifically those whose coeff. are zero, can't be represented)

AIMT-1 MSQ

Que: Let V_i's be vectors in R^n for $i = 1, 2, 3, 4$, then which of the following are correct?

a) if $\{V_1, V_2, V_3\}$ is linearly dependent, so is $\{V_1, V_2\}$

$\Rightarrow$ false, Not always.

b) if V_4 is not a linear combination of $\{V_1, V_2, V_3\}$ then $\{V_1, V_2, V_3, V_4\}$ is linearly independent.

$\Rightarrow$ false, not true if $\{V_1, V_2, V_3, V_4\}$ are linearly dependent and V_4 have 0 as coefficient in linear combination.

c) any set containing the zero vector is LD.

$\Rightarrow$ True

d) imp if $\{V_1, V_2, V_3\}$ is LD, so is $\{V_1, V_2, V_3, V_4\}$

$\Rightarrow$ True, Always as we can add V_4 with 0 coeff in LC of $\{V_1, V_2, V_3\}$

Trivial ↔ Obvious

★ **Linear Independence:** A set of vectors are LI if they are NOT LD.

★ if $c_1v_1 + c_2v_2 + c_3v_3 \ldots + c_nv_n = 0$
then if we have atleast one non zero c_i
then

$$v_i = \frac{-1}{c_i}(c_1v_1 + c_2v_2 \ldots + c_nv_n)$$

and hence it becomes LD, because we get LC for v_i.

Or if we cannot have any non zero c_i, then that set is LI. (only trivial soln)

Que: A set of vectors satisfies the below equation
$c_1v_1 + c_2v_2 \ldots + c_nv_n = 0$ for all $c_i = 0$
then $\{v_1, v_2 \ldots v_n\}$ is LI. T/F?

⇒ It seems to be true but the above statement is not giving any info. it will also be true for LD. so cannot say it is true. **FALSE.**

If the above eq was only true when all c_i are zero then it was true.

→ zero se aaega is trivial but sirf zero se aaega makes it LI.

→ If we are given three vectors' set and are asked to check if they are LI or LD, then we can hit and try and if we are able to get LC then it is LD else we don't know as we are doing just hit & trial.

☆ Always

$$\left\{ \underset{u}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}, \underset{v}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}, \underset{w}{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}} \right\}$$

It is LD as

$$\alpha u + \beta v = w$$

Here u, v are called convenient vectors.

☆ Vector Space: Collection of ~~space~~ vectors (zero vector is always required).

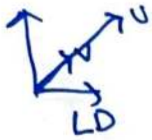
Example: R^2, R^3, R^4

☆ Filling the space (span): creating all vectors.

$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \& \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ are filling the space of ~~vector~~ R^2 as we can create any LC using them.

Do we have any other such set which can create LC for any vector?

⇒ Yes.



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Imp

☆ Any 2 linearly independent vectors can fill the space. i.e they can make Linear Combination of any vector.

Proof: Ex: $\alpha\begin{bmatrix}1\\2\end{bmatrix} + \beta\begin{bmatrix}2\\5\end{bmatrix} = \begin{bmatrix}3\\4\end{bmatrix}$

$$\left.\begin{aligned}\alpha + 2\beta &= 3\\ 2\alpha + 5\beta &= 4\end{aligned}\right\} \text{Can be solved easily.}$$

Also $\begin{bmatrix}1\\0\end{bmatrix}$ & $\begin{bmatrix}0\\1\end{bmatrix}$ are linearly independent & that's why they are able to fill the space of R^2.

Obviously they should be LI as if they will be not then equation will not be solved (use brain).

→ By 'filling R^2' we mean give any vector we can represent it using the filler vectors.

→ Why they (LIV) are able to fill R^2?
↳ will be discussed Later.

☆ So Two LDV can't fill the space i.e. will not be able to get LC of all vectors.

In case of set of one vector LI, because $cv=0$ is only true when $c=0$.
So it is LI. If that one vector is zero vector then it is LD as $cv=0$ is true for all c.

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Que: Does these set of vectors fill the space of R^2?

i) $\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ → Yes as LI

ii) $\left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 6 \\ 13 \end{bmatrix} \right\}$ → Yes as LI

iii) $\left\{ \underset{u}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}, \underset{v}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}, \underset{w}{\begin{bmatrix} 3 \\ 5 \end{bmatrix}} \right\}$ → Yes as two are LI & we can take 0 as coeff of 3^{rd}.

However w is redundant & most efficient (also called basis) of R^2 are $\{u, v\}$.

Que: Whether given set is LD or LI.

$\left\{ \underset{u}{\begin{bmatrix} 1 \\ 9 \end{bmatrix}}, \underset{v}{\begin{bmatrix} 5 \\ 3 \end{bmatrix}}, \underset{w}{\begin{bmatrix} 2 \\ 4 \end{bmatrix}} \right\}$

⇒ Sol^n: as u & v are LI and we know they can fill the space, so they can create w using some coeff so the set is LD.

So from above sol^n we can infer that, we cannot have more than 2 independent vectors in R^2.

* So definition of LD says that if in any set if atleast one vector can be represented as LC of other then it is LD ie from prev. notes any two LI vectors can represent LC of any vector in R^2. So we can say that any set ~~having atleast two~~ can't have more than two independent vectors in R^2. And any set having one pair of Independent vectors and then any other vector is LD.

Now similarly:

* We cannot have more than 3 LI vector in R^3.

* We cannot have more than n LI vector in R^n.

* Any two LI vectors in R^2 fill the space of R^2.

* Any 3 LI vectors in R^3 fill the space of R^3.

Conclusion:

1) If you have 2 vectors than you can easily check if they are LD or LI. (others can also be checked by rank - taught later)

2) If a set contains zero vector then it is LD set.

3). 5 LI vectors in R^5 can fill the space of R^5.
& so 6 vectors in R^5 can never be linearly I.

Que: Check if the following pairs of vectors are LI.
(No formal method - use intuition)

a) $\begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \end{bmatrix} \Rightarrow$ LD

b) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow$ LI

c) $\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \Rightarrow$ LI

Que: Determine if the following set of vectors are LI. Justify

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

As all are R^3 and R^3 cannot have more than 3 LI vectors. So set is LD.
(i.e. ☆ if they have more than n vectors in R^n then surely it is LD)

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To fill space
↕
To span space

Que: If A is a 3x5 matrix then the columns of A are LD

As: True, (5 vectors in R^3 are obviously LD).

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

Definition

Que: If a set contains more vectors than there are enteries in each vector, then the set is LD. i.e. any set $\{v_1, v_2, v_3 \ldots v_p\}$ in R^n is LD if $p > n$.

Ans True

Que: If there are 6 LD vectors in R^6, then these vectors cannot fill R^6.

Ans True

Que: Which is sufficient to fill R^6:

a) 6 LD

b) 5 LI

c) 7 any

d) 6 LI ✓ (~~Atleast~~ only requirement any addition will work)

$R^1 \rightarrow$ Scaler.

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★ **Summary till Now**

$\rightarrow$ $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$

Can you give me atleast one $c_i \neq 0$

Yes $\downarrow$ LD

No $\downarrow$ LI

★ Matrix X Vector

Ex:1

$$\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Also we can do.

$$1 \times \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Ex:2

$$\begin{bmatrix} 4 & 7 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow$$

$$1 \begin{bmatrix} 4 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 18 \\ 6 \end{bmatrix}$$

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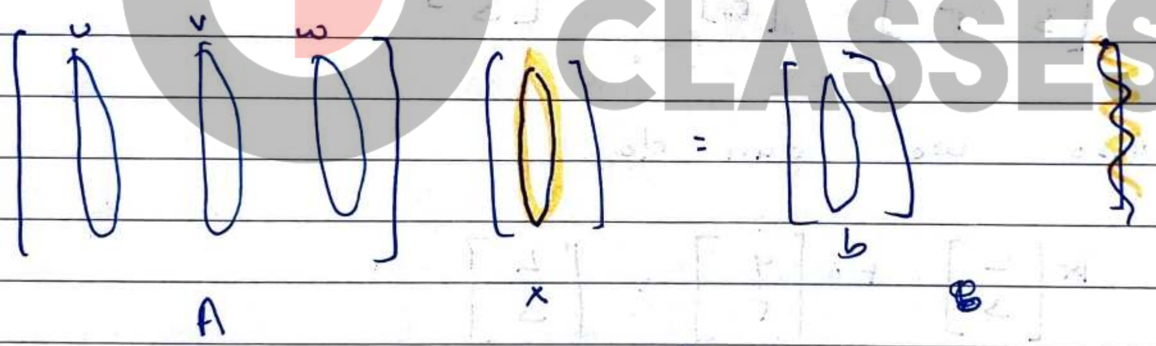
Ex:3

$$\begin{bmatrix} 1 & 5 & 9 \\ 2 & 6 & 10 \\ 3 & 7 & 11 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow$$

By L.C.

$$= 1\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0\begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix} + 1\begin{bmatrix} 9 \\ 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \\ 14 \end{bmatrix}$$

$Ax = b$



$\Rightarrow$ b is L.C. of columns of A with coeff as x.

Now we know to calculate b given A & x.

Find x.

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

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if b is → then x is

Eg: 1

$$\begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

2

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

3

$$\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

4

$$\begin{bmatrix} 6 \\ 9 \\ 12 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Just by → intusion

But Need some method

Que: If $Ax=0$ has some non trivial solution, then columns of A are LD.

⇒ True (can't guess it? Refer old notes again)

$$\underbrace{\begin{bmatrix} \uparrow & \uparrow & \uparrow \\ a_1 & a_2 & a_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{0}$$

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So we can write it as:

$$c_1 \begin{bmatrix} \uparrow \\ a_1 \\ \downarrow \end{bmatrix} + c_2 \begin{bmatrix} \uparrow \\ a_2 \\ \downarrow \end{bmatrix} + c_3 \begin{bmatrix} \uparrow \\ a_3 \\ \downarrow \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow$ As sol^n is nontrivial then we have any $c \neq 0$
So we can write one C as l.c. of other
So it is LD.

☆ We can get b, given A & x directly, but getting x given A & b is what we call Solving system of linear equations.

$x - 3y = 5$ (var.)
$2x + 4y = 10$ (coeff.)
} system of linear equation

these are linear equations because, neither their ~~coeff.~~ variables have power (x^2, y^3, etc) nor multiplicative (xy, x^2y, etc)

Some non Linear eq. are:
$x^2 + y = 5$, $x - xy = 5$, $-\sqrt{x} + y = 41$

In general, given a set of lin. eq. we want to solve for values of variables

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For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking here).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click here to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal